

UNIT-2 Coding question

1. Taking Input on Coding Platforms

(a) Single Value in a Single Line

- **Print Hello World** – First basic input/output

GfG Problem:

<https://www.geeksforgeeks.org/dsa/hello-world-program-first-program-while-learning-programming/>

- <https://leetcode.com/problems/single-number/description/>
- <https://leetcode.com/problems/single-number-iii/description>
- <https://leetcode.com/problems/missing-number/description/>

(b) Multiple Known Values in a Single Line

- **Add Two Numbers** – Read two integers in one line and print their sum.

GfG Problem: <https://www.geeksforgeeks.org/problems/add-two-numbers/0>

- <https://www.geeksforgeeks.org/find-two-numbers-odd-occurrences-array>
- <https://www.hackerrank.com/challenges/lonely-integer>
- <https://www.geeksforgeeks.org/find-elements-frequency-1-array>

(c) Multiple Known Values in Multiple Lines

- **Sum of Numbers from Multiple Lines** – Read multiple lines of input and compute the total sum.

HackerRank – Basic I/O <https://www.hackerrank.com/challenges/cpp-input-and-output/problem>

- <https://leetcode.com/problems/find-all-numbers-disappeared-in-an-array/description>
- <https://leetcode.com/problems/xor-queries-of-a-subarray/description/>

2. Operators (Arithmetic, Relational, Logical, Conditional, Bitwise)

Arithmetic Operators

- **Add, Subtract, Multiply** – Basic arithmetic operations.

<https://www.geeksforgeeks.org/problems/arithmetic-operators-1605594922>

- <https://www.hackerrank.com/challenges/a-very-big-sum/problem>
- https://leetcode.com/problems/plus-one/description/?utm_source=chatgpt.com

Relational & Logical Operators

- **Relational Operators** – Compare two integers.
HackerRank Problem: <https://www.hackerrank.com/challenges/bash-tutorials---comparing-numbers/problem>
- <https://www.hackerrank.com/challenges/compare-the-triplets/>
- <https://leetcode.com/problems/valid-triangle-number/description/>

Conditional (Ternary) Operator

- **Find Maximum of Two Numbers Using Ternary Operator**
Problem: <https://www.geeksforgeeks.org/dsa/program-to-find-the-largest-number-using-ternary-operator/>
- <https://www.hackerrank.com/challenges/grading/problem>
- <https://leetcode.com/problems/max-consecutive-ones/description>

Bitwise Operators

- **Bit Manipulation** – Find if **k**th bit is set.
GfG Problem : <https://www.geeksforgeeks.org/problems/bit-manipulation-1666686020/0>
- <https://www.hackerrank.com/challenges/flipping-bits/problem>

3. Program Flow Control (if, if-else, nested if-else, else-if ladder, switch-case)

Simple if and if-else

- Problem: <https://www.geeksforgeeks.org/problems/odd-or-even3618/1>
- <https://www.geeksforgeeks.org/problems/java-if-else-decision-making0924/>
- <https://www.hackerrank.com/challenges/write-a-function/problem>
- <https://www.geeksforgeeks.org/problems/vowel-or-not0831/1>
- <https://www.hackerrank.com/challenges/grading/problem>
- <https://www.hackerrank.com/challenges/30-loops/problem>

Nested if-else / else-if Ladder

- <https://www.hackerrank.com/challenges/c-tutorial-conditional-if-else/problem>
- <https://www.geeksforgeeks.org/problems/java-if-else-decision-making0924/1>
- <https://www.geeksforgeeks.org/dsa/program-to-check-whether-a-triangle-is-equilateral-isosceles-or-scalene/>

Switch-case

- <https://www.geeksforgeeks.org/problems/c-switch-case-statement5900/1>
- <https://leetcode.com/problems/integer-to-roman/description/>
- <https://leetcode.com/problems/fizz-buzz/description/>

Math library question:

1. Square Root and Power

- <https://leetcode.com/problems/valid-perfect-square/description/>
- <https://leetcode.com/problems/sqrtx/description/>
- <https://leetcode.com/problems/power-of-three/description/>

2. Absolute Value and Rounding

- <https://www.hackerrank.com/challenges/30-operators/problem>
- <https://www.geeksforgeeks.org/problems/distance-between-two-points3202>
- <https://www.hackerrank.com/challenges/compare-the-triplets/problem>

3. Trigonometry and Angles

- <https://www.geeksforgeeks.org/problems/angle-between-hour-and-minute-hand0545/1>
- <https://www.hackerrank.com/domains/mathematics>

4. Logarithms and Exponents

- <https://leetcode.com/problems/power-of-two/description/>
- <https://leetcode.com/problems/super-pow/description/>

5. Rounding and Ceil/Floor

- <https://www.geeksforgeeks.org/problems/ceil-the-floor2802/1>